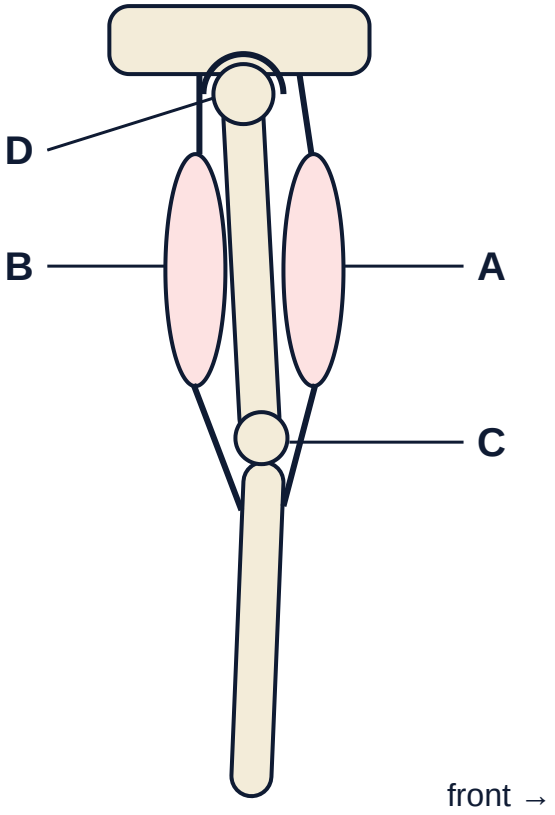
Stage 8 Science — Practice Test 1 — Mark Scheme

Paper 1: 45 minutes · calculator allowed · Paper 2: 45 minutes · Total 100 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 50 marks

1 On sports day, Salma takes part in the long jump. The diagram shows the bones and two muscles of her leg, seen from the side. The front of the leg is on the right. **[1]**



(a) Which letter, A, B, C or D, shows a hinge joint?

Answer: C
Notes / working: C is the knee. The knee is a hinge joint: it lets the leg bend and straighten in one direction only, like a door hinge. D is the hip, which is a ball-and-socket joint.

(b) Name the type of joint at D. **[1]**

Answer: ball and socket or ball-and-socket or ball and socket joint or ball-and-socket joint or a ball and socket joint or a ball-and-socket joint
Notes / working: D is the hip. The round end (ball) of the thigh bone fits into a cup (socket) in the hip bone, so the leg can move in many directions and turn round: a **ball-and-socket joint**.

(c) Salma straightens her leg at the knee to push off the ground. Which muscle, A or B, contracts to do this? **[1]**

> **Answer:** A
> **Notes / working:** To straighten the knee, the bone below the knee must be pulled forwards. Muscle A is on the front of the thigh and its tendon is fixed to the front of the lower leg bone, so **A** contracts (and B relaxes).

(d) Muscles A and B are called an antagonistic pair. Explain what this means and why muscles must work in pairs. **[2]**

> **Answer:** 1 mark: the two muscles have opposite effects / when one contracts the other relaxes (e.g. A straightens the leg, B bends it). 1 mark: a muscle can only pull (contract); it cannot push, so a second muscle is needed to move the bone back. Do not accept: 'muscles push the bones'.
> **Notes / working:** A muscle can only **pull** when it contracts; it cannot push. So one muscle (A) pulls the lower leg straight and another muscle (B) is needed to pull it back to bend the knee. When one contracts, the other relaxes. Two muscles that work against each other like this are an antagonistic pair.

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2 Youssef runs an 800 m race on sports day. His teacher counts his breathing rate before and after the race. **[1]**

Time	Breathing rate in breaths per minute
before the race	15
1 minute after the race	42
5 minutes after the race	24
10 minutes after the race	16

Use the table to answer the questions.

(a) By how many breaths per minute did Youssef's breathing rate increase from before the race to 1 minute after the race?

> **Answer:** 27 breaths per minute
> **Notes / working:** 42 − 15 = **27** breaths per minute.

(b) Which describes what happens when Youssef breathes in? **[1]**

- ☐ A) The diaphragm relaxes and moves up; the ribs move down and in.
- ☐ B) The diaphragm contracts and moves down; the ribs move up and out.
- ☐ C) The diaphragm contracts and moves up; the ribs move down and in.
- ☐ D) The diaphragm relaxes and moves down; the ribs move up and out.

Answer: B) The diaphragm contracts and moves down; the ribs move up and out.
Notes / working: Breathing in: the diaphragm **contracts and flattens (moves down)** and the intercostal muscles contract to pull the ribs **up and out**. The chest gets bigger, the pressure inside falls and air rushes in. The answer is B.

(c) Explain why Youssef's breathing rate is much higher just after the race. Use the word **respiration** in your answer. **[2]**

Answer: 1 mark: his muscles were respiring faster / needed more energy (from respiration) to run. 1 mark: so more oxygen must be taken in / more carbon dioxide must be removed (from the blood). Accept: to repay the oxygen his muscles used. Do not accept 'to make energy' without respiration or oxygen.
Notes / working: Running muscles need a lot of energy. The energy is released by **aerobic respiration**, which uses oxygen and makes carbon dioxide. So Youssef breathes faster to take in more oxygen for his muscles and to get rid of the extra carbon dioxide.

(d) Youssef says: "Breathing and respiration are the same thing." Explain why he is wrong. **[2]**

Answer: 1 mark: breathing is moving air into and out of the lungs (using the diaphragm and intercostal muscles). 1 mark: respiration is a chemical reaction in (every) cell / in mitochondria that releases energy from glucose (using oxygen). Both ideas are needed for 2 marks.
Notes / working: Breathing is a physical process: muscles move air into and out of the lungs. **Respiration** is a chemical reaction that happens inside every living cell (in the mitochondria): glucose reacts with oxygen to release energy.

(e) The word equation for aerobic respiration is: **[1]**
glucose + oxygen → _____ + water
Write the missing product.

Answer: carbon dioxide or CO2 or CO_2 or CO_2
Notes / working: glucose + oxygen → **carbon dioxide** + water.

(f) Describe how oxygen gets from the air in the alveoli into Youssef's blood. **[2]**

Answer: 1 mark: by diffusion. 1 mark: from a high concentration (of oxygen) in the alveoli to a low concentration in the blood / through the thin walls of the alveoli and capillaries (one cell thick). Do not accept 'oxygen is pumped' or 'sucked'.
Notes / working: Oxygen moves by **diffusion**. There is a high concentration of oxygen in the air in the alveoli and a low concentration in the blood arriving at the lungs, so oxygen moves across the very thin walls of the alveoli and the capillaries into the blood.

(g) Which statement about aerobic respiration is correct? **[1]**

- ☐ A) It happens only in animal cells, inside the nucleus.
- ☐ B) It happens only in the lungs, when air is breathed in.
- ☐ C) It happens in the mitochondria of plant and animal cells and releases energy in a controlled way.
- ☐ D) It releases all the energy in glucose at once, like burning.

Answer: C) It happens in the mitochondria of plant and animal cells and releases energy in a controlled way.
Notes / working: Aerobic respiration happens in the **mitochondria** of both plant and animal cells. It releases energy in small, **controlled** steps, not all at once as in burning. The answer is C.

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3 Egypt has built a very large solar park in the desert near Aswan. Egypt also gets electricity from wind farms and from power stations that burn natural gas. **[1]**

(a) Which list contains **only** renewable energy resources?

- ☐ A) wind, solar, natural gas
- ☐ B) tidal, coal, wind
- ☐ C) solar, wind, tidal
- ☐ D) oil, natural gas, coal

Answer: C) solar, wind, tidal
Notes / working: Solar, wind and tidal power will not run out, so they are renewable. Coal, oil and natural gas are fossil fuels, which are non-renewable. The answer is C.

(b) Suggest **one** reason why the desert in southern Egypt is a good place for a solar park. **[1]**

Answer: 1 mark for any one: a lot of sunshine / many sunny days / clear skies with little cloud / strong sunlight; large areas of empty land that are not used for farming or homes. Do not accept 'it is hot' on its own.
Notes / working: Solar panels need sunlight. Southern Egypt has strong sunshine on almost every day of the year, with very few clouds. There is also a lot of empty desert land, so the panels do not take land needed for farms or towns.

(c) A company now makes water bottles from bioplastic instead of plastic made from crude oil. **[1]**
Which is an advantage of bioplastic?

- ☐ A) It is made from plants, which can be grown again, so it is renewable.
- ☐ B) It is made from coal, so it lasts for millions of years.
- ☐ C) It is made from crude oil, so it is cheaper.
- ☐ D) It can never break down, so it does not need recycling.

Answer: A) It is made from plants, which can be grown again, so it is renewable.
Notes / working: Bioplastics are made from plant materials (for example starch from maize or sugar cane). The plants can be grown again every year, so bioplastic is a **renewable** resource, while crude oil will run out. The answer is A.

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4 Mariam investigates how four metals, W, X, Y and Z, react with dilute hydrochloric acid. The metals are copper, iron, magnesium and zinc, in a random order. **[1]**
She adds 0.5 g of each metal powder to 20 cm^3 of the acid and measures the volume of gas made in 1 minute. She does each test three times.

Metal	Test 1 in cm^3	Test 2 in cm^3	Test 3 in cm^3
W	6	5	7
X	0	0	0
Y	52	50	51
Z	21	35	23

Use the results in the table to answer the questions.

(a) Write the letters of the metals in order of reactivity, starting with the **most** reactive.

Answer: YZWX or Y Z W X or Y, Z, W, X or Y,Z,W,X or Y>Z>W>X or Y > Z > W > X or Y-Z-W-X or Y Z W X
Notes / working: The more reactive the metal, the more gas it makes in 1 minute. Means: Y about 51, Z about 22, W 6, X 0. Order: **Y, Z, W, X**.

(b) Which metal, W, X, Y or Z, is iron? **[1]**

Answer: W
Notes / working: Reactivity order: magnesium > zinc > iron > copper. So Y = magnesium, Z = zinc, **W = iron** and X = copper (copper does not react with dilute acid).

(c) Which piece of equipment should Mariam use to measure the volume of gas? **[1]**

- ☐ A) a thermometer
- ☐ B) a top-pan balance
- ☐ C) a stopwatch
- ☐ D) a gas syringe

Answer: D) a gas syringe
Notes / working: A **gas syringe** collects the gas and has a scale in cm^3. The stopwatch times the 1 minute but does not measure volume. The answer is D.

(d) One result in the table is anomalous. Write down this result. **[1]**

Answer: 35 cm^3
Notes / working: Metal Z gave 21 and 23 cm^3, but test 2 gave **35 cm^3**. This does not fit the pattern of the other results, so it is anomalous.

(e) Calculate the mean volume of gas for metal Z. Do not include the anomalous result. **[1]**

Answer: 22 cm^3
Notes / working: (21 + 23) ÷ 2 = 44 ÷ 2 = **22 cm^3**. (Including 35 would give 26.3 cm^3, which is wrong.)

(f) Complete the word equation for the reaction of iron with hydrochloric acid. **[1]**
iron + hydrochloric acid → _____ + hydrogen

Answer: iron chloride or iron(II) chloride or iron (II) chloride or iron(ii) chloride or iron 2 chloride
Notes / working: metal + hydrochloric acid → metal **chloride** + hydrogen, so iron + hydrochloric acid → **iron chloride** + hydrogen.

(g) Mariam's teacher puts a small piece of gold into the same acid. Predict what Mariam will see, and explain your prediction. **[2]**

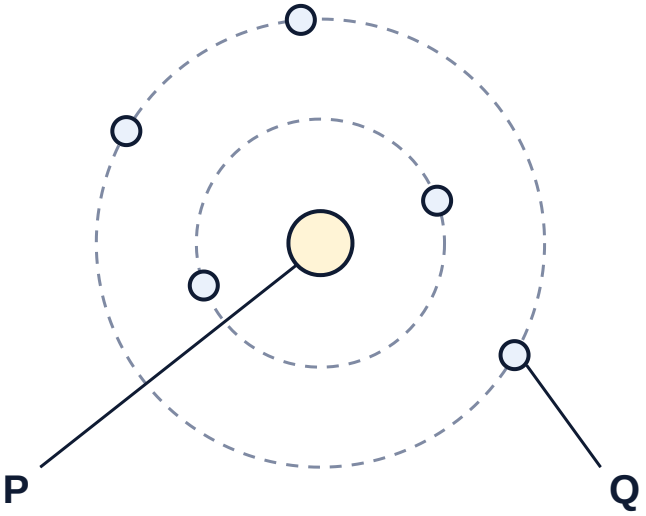
Answer: 1 mark: no bubbles / no gas / no reaction / no change. 1 mark: gold is (very) unreactive / inert / less reactive than copper (and copper did not react). Do not accept 'gold dissolves slowly'.
Notes / working: Nothing happens: no bubbles of gas. Gold is at the very bottom of the reactivity series, below copper, which did not react either. Gold is so unreactive that it is described as **inert**.

(h) Dilute hydrochloric acid is an irritant. Describe **one** way Mariam can reduce the risk of harm. **[1]**

> **Answer:** 1 mark for any one: wear safety glasses / goggles (to protect the eyes); wash acid off skin with plenty of water at once; use small volumes / dilute acid; stand up and keep the beaker away from the edge of the bench; wipe up spills straight away. Do not accept 'be careful' on its own.
> **Notes / working:** For example: wear **safety goggles** so the acid cannot splash into her eyes, and wash any acid off her skin with plenty of cold water.

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5 The diagram shows the Rutherford model of an atom. **[1]**



(a) Which letter, P or Q, shows a particle with a **negative** charge?

> **Answer:** Q
> **Notes / working:** Q is an electron moving around the nucleus. Electrons have a **negative** charge. P is the nucleus, which is positive.

(b) Which row gives the correct charges of a proton and a neutron? **[1]**

Row	proton	neutron
A	positive	no charge
B	negative	positive
C	no charge	negative
D	positive	negative

Choose the correct row.

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

Answer: A) A
Notes / working: Protons are **positive** and neutrons have **no charge** (they are neutral). Electrons are negative. The answer is A.

(c) An atom of fluorine has 9 electrons and 10 neutrons. How many protons does it have? **[1]**

Answer: 9
Notes / working: An atom has no overall charge, so the number of protons (+) equals the number of electrons (−): **9 protons**. The 10 neutrons have no charge and do not change this.

(d) Explain what holds the electrons in an atom. **[2]**

Answer: 1 mark: electrostatic attraction / the attraction between opposite charges. 1 mark: between the negative electrons and the positive nucleus / positive protons. Do not accept 'gravity' or 'magnetism'.
Notes / working: The electrons are negative and the protons in the nucleus are positive. Opposite charges attract, so there is an **electrostatic attraction** between the electrons and the nucleus. This force holds the atom together.

(e) Which statement describes the Rutherford model of the atom? **[1]**

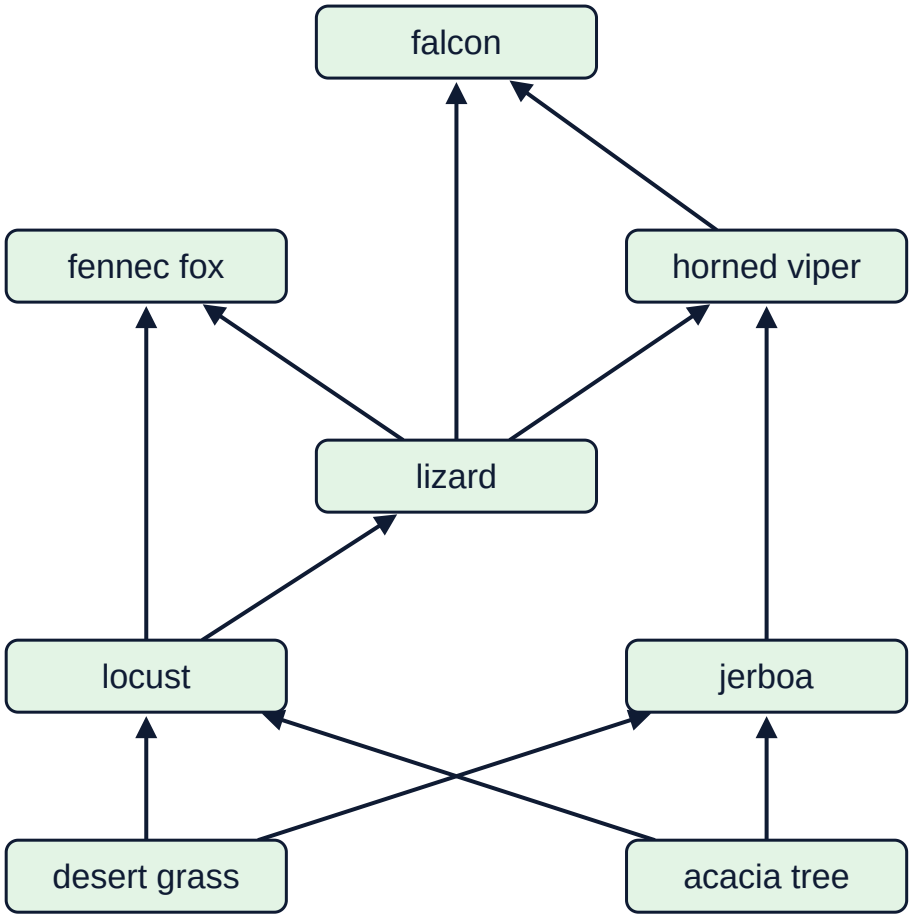
- ☐ A) Positive charge is spread through a ball, with electrons dotted inside it like fruit in a cake.
- ☐ B) A tiny positive nucleus holds almost all the mass; electrons move around it, and most of the atom is empty space.
- ☐ C) An atom is a solid ball with no smaller particles inside it.
- ☐ D) A large negative nucleus has positive electrons moving around it.

Answer: B) A tiny positive nucleus holds almost all the mass; electrons move around it, and most of the atom is empty space.
Notes / working: Rutherford's model has a tiny, dense, **positive nucleus** with nearly all the mass, electrons moving around it, and mostly **empty space**. A is the older plum-pudding model. The answer is B.

6 The Western Desert of Egypt is an ecosystem. The diagram shows part of a desert food web. **[1]**
Farmers near an oasis spray a pesticide that is not broken down in living things. Scientists measure the pesticide in one food chain.

Organism	desert grass	locust	lizard	horned viper	falcon
Pesticide in parts per million	0.05	0.6	4.5	18	72

Use the food web and the table to answer the questions.



(a) Which of these is a **habitat** within the desert ecosystem?

- ☐ A) the hot, dry climate of the desert
- ☐ B) all the animals that live in the desert
- ☐ C) a small oasis pool with palm trees
- ☐ D) the food web in the diagram

Answer: C) a small oasis pool with palm trees
Notes / working: A habitat is the place where an organism lives. An oasis pool is one of many habitats in the desert ecosystem (others are sand dunes, rocky slopes and burrows). The climate, the animals and the food web are not places. The answer is C.

(b) Name **one** producer in this food web. **[1]**

Answer: desert grass or grass or acacia tree or acacia or acacia trees
Notes / working: Producers are plants that make their own food by photosynthesis: **desert grass** or **acacia tree**.

(c) Name the animal that competes with the lizard for locusts. **[1]**

Answer: fennec fox or fennec foxes or fox or the fennec fox
Notes / working: Follow the arrows from the locust: they go to the lizard and to the **fennec fox**. Both eat locusts, so they compete for them.

(d) How many times greater is the concentration of pesticide in the falcon than in the locust? **[1]**

Answer: 120
Notes / working: 72 ÷ 0.6 = **120** times greater.

(e) Explain why the concentration of pesticide is highest in the falcon. **[2]**

Answer: 1 mark: the pesticide is not broken down / not excreted, so it stays (is stored) in the body of each animal. 1 mark: each animal eats many of the organisms below it, so the pesticide builds up (accumulates) at each step / the falcon is at the top of the food chain. Accept 'bioaccumulation' with an explanation.
Notes / working: The pesticide is not broken down or removed, so it stays in the animals' bodies. A lizard eats many locusts, a viper eats many lizards and a falcon eats many vipers, so at each step the pesticide becomes more concentrated (**bioaccumulation**). The falcon is at the top of the chain, so it has the most.

(f) Trucks bring a new species of rat to the oasis by accident. The rat breeds quickly and eats seeds and grass. Suggest **one** way the rat could make the number of jerboas fall. **[1]**

Answer: 1 mark for any one: the rat competes with the jerboas for the same food (seeds / grass / plants); it competes for space / burrows / shelter; it brings a new disease that the jerboas cannot fight. The answer must include a reason, not just 'jerboas die'.
Notes / working: The rat eats the same food as the jerboa (seeds and grass). It breeds quickly, so it can eat most of the food and the jerboas cannot get enough. This **competition** makes the number of jerboas fall.

7 Athletes need a healthy diet and healthy blood and lungs. **[1]**

(a) Which row shows a good source of vitamin C and one of its functions?

- ☐ A) fish oil — helps the body take in calcium for strong bones
- ☐ B) carrots — needed for good eyesight in dim light
- ☐ C) red meat — needed to make haemoglobin in red blood cells
- ☐ D) oranges and guava — keep the skin and gums healthy

Answer: D) oranges and guava — keep the skin and gums healthy
Notes / working: Citrus fruits and guava are rich in **vitamin C**, which keeps skin and gums healthy and helps wounds heal. A describes vitamin D, B vitamin A and C the mineral iron. The answer is D.

(b) The night before a long race, Karim eats a big plate of pasta. Pasta is rich in one nutrient that the body uses mainly as a source of energy. Name this nutrient. **[1]**

Answer: carbohydrate or carbohydrates or starch or carbohydrate (starch)
Notes / working: Pasta is rich in **carbohydrate** (starch). Carbohydrates are broken down to glucose, which is used in respiration to release energy; extra can be stored for the race.

(c) Name the part of the blood that carries glucose from the small intestine to the muscles. **[1]**

Answer: plasma or blood plasma or the plasma
Notes / working: Dissolved nutrients such as glucose, and carbon dioxide, are carried in the **plasma**, the liquid part of the blood.

(d) Explain why a person who smokes usually cannot run as far as a person who does not smoke. **[2]**

Answer: 1 mark for any one cause: carbon monoxide (in smoke) is carried by red blood cells instead of oxygen / reduces the oxygen the blood carries; OR tar / smoke damages the alveoli, so less gas exchange; OR smoke damages the airways / makes mucus. 1 mark: so less oxygen reaches the muscles, so less (aerobic) respiration / less energy is released.
Notes / working: Cigarette smoke contains **carbon monoxide**, which is carried by red blood cells in place of oxygen, and tar, which damages the alveoli so less gas exchange happens. Less oxygen reaches the muscles, so they release less energy by aerobic respiration and the smoker tires quickly.

8 This question is about the Earth and space. **[1]**

(a) Which description of a galaxy is correct?

- ☐ A) a huge group of stars and planetary systems, with dust and gas, held together by gravity
- ☐ B) one star with the planets that orbit it
- ☐ C) a cloud of gas around one planet
- ☐ D) all the moons in the Solar System

Answer: A) a huge group of stars and planetary systems, with dust and gas, held together by gravity
Notes / working: A **galaxy** contains billions of stars, many with planetary systems, plus clouds of dust and gas, all held together by gravity. B describes a planetary system, such as our Solar System. The answer is A.

(b) Which statement about asteroids is correct? **[1]**

- ☐ A) Asteroids are small stars that give out their own light.
- ☐ B) Asteroids are rocks, smaller than planets, left over from when the planetary system formed.
- ☐ C) Asteroids are clouds of gas between the planets.
- ☐ D) Asteroids are bigger than planets and orbit the Moon.

Answer: B) Asteroids are rocks, smaller than planets, left over from when the planetary system formed.
Notes / working: Asteroids are **rocks**, smaller than planets. They are pieces of rock **left over from the formation** of the Solar System that never joined together to make a planet. The answer is B.

(c) Which statement describes the **climate** of Aswan, not its weather? **[1]**

- ☐ A) It rained for ten minutes in Aswan yesterday.
- ☐ B) Tomorrow the temperature in Aswan will be 41 °C.
- ☐ C) There is a strong wind in Aswan this morning.
- ☐ D) Aswan has very hot, dry summers and almost no rain in most years.

Answer: D) Aswan has very hot, dry summers and almost no rain in most years.
Notes / working: Weather is the conditions on one day or over a few days. **Climate** is the usual (average) pattern of weather in a place over many years. Only D describes the pattern over many years. The answer is D.

(d) Scientists say that the Earth's climate has changed many times between ice ages and warm periods. Give **one** piece of evidence scientists use for this. **[1]**

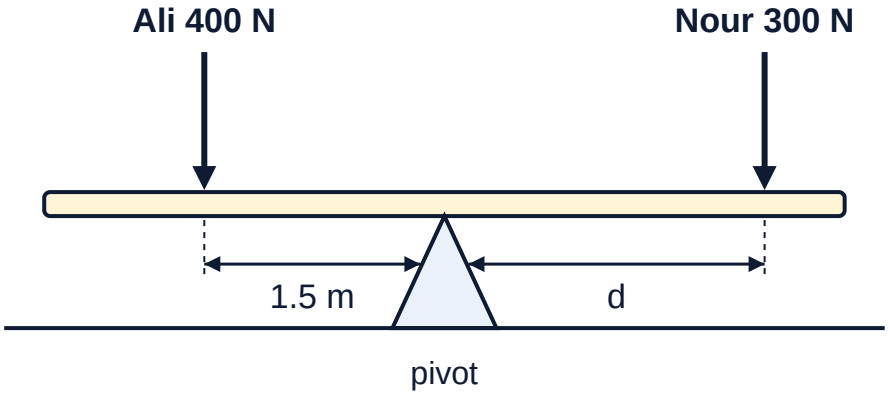
Answer: 1 mark for any one: ice cores (the air bubbles / layers trapped in ancient ice); fossils of plants or animals that lived in a different climate (e.g. tropical fossils in cold places); rocks and land shapes made by glaciers in places that are warm today; layers of sediment on the sea floor. Do not accept 'the weather changes every day'.
Notes / working: For example, **ice cores**: deep ice in Antarctica contains tiny bubbles of air trapped hundreds of thousands of years ago, which show what the atmosphere and temperature were like. Fossils of animals or plants from a very different climate are also evidence.

(e) Name **one** gas in the atmosphere whose increase is making the Earth's climate warmer. **[1]**

Answer: carbon dioxide or CO2 or CO_2 or CO_2 or methane or CH4 or CH_4 or CH_4 or water vapour or nitrous oxide
Notes / working: Carbon dioxide (or methane). These greenhouse gases trap heat in the atmosphere; more of them makes the Earth's average temperature rise.

Paper 2 — 45 minutes — 50 marks

1 Ali and Nour sit on a seesaw in the park. The seesaw is balanced. Ali weighs 400 N and sits 1.5 m from the pivot. Nour weighs 300 N and sits a distance d from the pivot. **[2]**



(a) Calculate the moment of Ali's weight about the pivot. Show your working.

Answer: 600 N m
Notes / working: moment = force × distance = 400 × 1.5 = **600 N m**. (M1 for 400 × 1.5 seen; A1 for 600 N m.)

(b) The seesaw is balanced. Calculate the distance d. Show your working. **[2]**

Answer: 2 or 2.0 m
Notes / working: Balanced, so clockwise moment = anticlockwise moment: 300 × d = 600, d = 600 ÷ 300 = **2.0 m**. (M1 for 600 ÷ 300 or 300 × d = 600; A1 for 2 m. ecf from (a).)

(c) Nour moves and sits 1.5 m from the pivot instead. What happens? **[1]**

- ☐ A) Ali's end goes down, because Ali's moment is now bigger than Nour's.
- ☐ B) Nour's end goes down, because Nour's moment is now bigger than Ali's.
- ☐ C) The seesaw stays balanced, because they are both 1.5 m from the pivot.
- ☐ D) The seesaw stays balanced, because the pivot is in the middle.

Answer: A) Ali's end goes down, because Ali's moment is now bigger than Nour's.
Notes / working: Nour's moment is now 300 × 1.5 = 450 N m, which is less than Ali's 600 N m. The moments are unbalanced, so the seesaw turns and **Ali's end goes down**. The answer is A.

(d) At home, Ali pushes on a door handle with a force of 15 N. The handle is 0.80 m from the hinges. Calculate the moment of Ali's push about the hinges. **[1]**

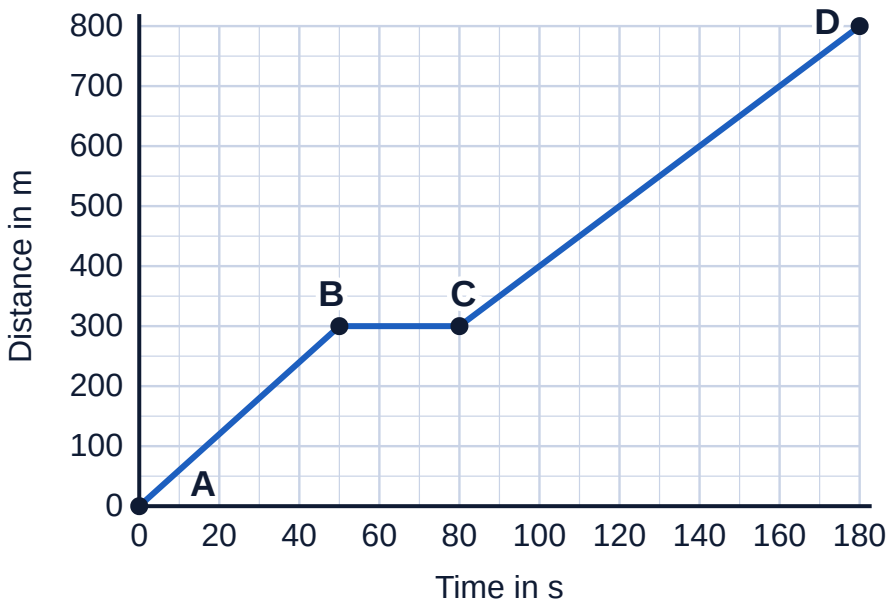
Answer: 12 N m
Notes / working: moment = 15 × 0.80 = **12 N m**.

(e) Explain why door handles are fixed on the side of the door furthest from the hinges. **[2]**

> **Answer:** 1 mark: the distance from the pivot (hinges) is larger. 1 mark: so the same force gives a bigger moment (turning effect) / a smaller force is needed to open the door (moment = force × distance). Do not accept 'it is easier' with no reason.
> **Notes / working:** The hinges are the pivot. moment = force × distance, so the larger the distance from the hinges, the **bigger the moment** for the same force. With the handle far from the hinges, only a small force is needed to open the door.

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2 Karim runs in an 800 m race on sports day. The distance–time graph shows his race. He stops at B to tie his shoelace, then runs again from C. **[1]**



(a) For how many seconds does Karim stop?

> **Answer:** 30 s
> **Notes / working:** The flat part BC goes from 50 s to 80 s: 80 − 50 = **30 s**. A flat (horizontal) line means the distance is not changing, so he is not moving.

(b) Calculate Karim's speed between C and D. Show your working. **[2]**

> **Answer:** 5 m/s
> **Notes / working:** Distance from C to D = 800 − 300 = 500 m. Time = 180 − 80 = 100 s. speed = distance ÷ time = 500 ÷ 100 = **5 m/s**. (M1 for 500 and 100 read correctly or 500 ÷ 100; A1 for 5.)

(c) Calculate Karim's average speed for the whole race. Give your answer to 1 decimal place. **[2]**

> **Answer:** 4.4 m/s
> **Notes / working:** average speed = total distance ÷ total time = 800 ÷ 180 = 4.444… = **4.4 m/s** (1 d.p.). The 30 s stop counts in the total time. (M1 for 800 ÷ 180; A1 for 4.4.)

(d) Between C and D Karim runs at a steady speed. Which statement about the forces on Karim is correct? **[1]**

- ☐ A) The forward force is bigger than the backward forces, so he speeds up.
- ☐ B) There are no forces acting on him.
- ☐ C) The forward force and the backward forces are balanced, so his speed does not change.
- ☐ D) The backward forces are bigger than the forward force, so he slows down.

> **Answer:** C) The forward force and the backward forces are balanced, so his speed does not change.
> **Notes / working:** CD is a straight sloping line, so the speed is constant. A constant speed means the forces are **balanced**: the forward force equals the air resistance and friction. The answer is C.

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3 This question is about pressure and particles on sports day. **[2]**

(a) A shot-put athlete weighs 720 N. The total area of the soles of both her shoes touching the ground is 0.045 m². Calculate the pressure she puts on the ground. Show your working.

> **Answer:** 16000 Pa
> **Notes / working:** pressure = force ÷ area = 720 ÷ 0.045 = **16 000 Pa** (N/m²). (M1 for 720 ÷ 0.045; A1 for 16 000.)

(b) Sprinters wear shoes with small metal spikes underneath. Why do the spikes help them to grip the track? **[1]**

- ☐ A) The spikes increase the area touching the track, so the pressure is smaller.
- ☐ B) The spikes have a small area, so the pressure is large and they press into the track.
- ☐ C) The spikes make the runner's weight smaller.
- ☐ D) The spikes have a large area, so the pressure is large.

> **Answer:** B) The spikes have a small area, so the pressure is large and they press into the track.
> **Notes / working:** The runner's weight acts on the tiny area of the spike tips. A small area gives a **large pressure**, so the spikes press into the track and grip it. The answer is B.

(c) A teacher pumps more air into a football. Use ideas about particles to explain why the pressure inside the ball increases. **[2]**

> **Answer:** 1 mark: there are more (air) particles inside the ball. 1 mark: so there are more collisions (of particles) with the inside wall of the ball each second / more force on the wall. Do not accept 'the particles get bigger' or 'the particles expand'.
> **Notes / working:** Gas pressure is caused by particles hitting the walls of the container. Pumping adds **more particles** to the same space, so there are **more collisions** with the inside of the ball every second, which push harder on it: the pressure goes up.

(d) Spectators at the far end of the field can smell the food being grilled. Use ideas about particles to explain how the smell reaches them. **[2]**

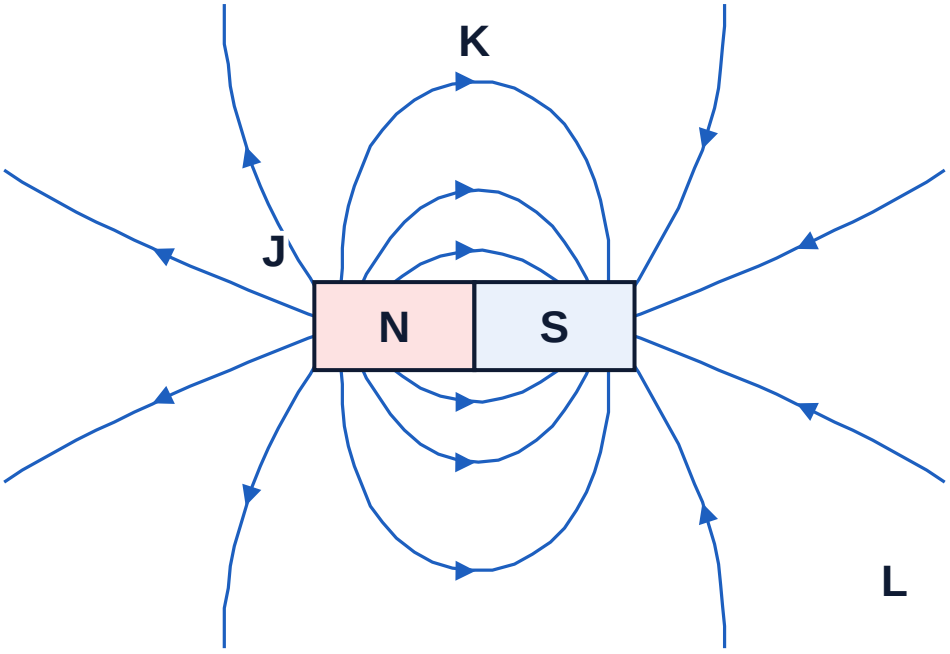
> **Answer:** 1 mark: the particles (of the smell / gas) move randomly / constantly / in all directions. 1 mark: they spread out from where there are many (high concentration) to where there are few (low concentration) / they mix (intermingle) with the air particles. Accept 'diffusion' as part of the second mark only with the idea of spreading or mixing.
> **Notes / working:** This is **diffusion**. Particles of the smelly gas are always moving randomly. They collide with and mix with the air particles and spread out from where they are concentrated (near the grill) to where there are few of them, until some reach the spectators' noses.

4 A scrapyard uses a crane with a large electromagnet to lift old cars. **[1]**

Sami makes a model of the crane. He winds 20 turns of insulated wire round an iron nail. He changes the current and counts how many small steel nails the electromagnet lifts.

Current in A	0.2	0.4	0.6	0.8	1.0
Number of steel nails lifted	3	6	9	5	15

Parts (e) to (j) are about Sami's investigation.



(a) The diagram shows the magnetic field around a bar magnet. At which point, J, K or L, is the magnetic field strongest?

Answer: J
Notes / working: The field is strongest where the field lines are **closest together**, which is near the poles. J is next to the N pole.

(b) Which statement about the magnetic field lines is correct? **[1]**

- ☐ A) Outside the magnet they go from the S pole to the N pole.
- ☐ B) They are furthest apart near the poles.
- ☐ C) They cross each other near the middle of the magnet.
- ☐ D) Outside the magnet they go from the N pole to the S pole.

Answer: D) Outside the magnet they go from the N pole to the S pole.
Notes / working: Field lines point from the **north pole to the south pole** outside the magnet (the arrows in the diagram). They never cross, and they are closest near the poles. The answer is D.

(c) Which of these is an electromagnet? **[1]**

- ☐ A) a steel nail stroked many times with a bar magnet
- ☐ B) a coil of insulated wire wrapped around an iron nail, with the wire connected to a cell
- ☐ C) an iron nail rubbed with a cloth
- ☐ D) a coil of insulated wire wrapped around an iron nail, with both ends of the wire not connected

Answer: B) a coil of insulated wire wrapped around an iron nail, with the wire connected to a cell
Notes / working: An electromagnet is a **coil of wire with a current flowing through it**, usually wound round an iron core. Without a current (D) there is no magnetic field. A makes a permanent magnet. The answer is B.

(d) Explain why the scrapyard uses an electromagnet and not a large permanent magnet. **[1]**

Answer: 1 mark: an electromagnet can be switched off (by turning off the current), so the scrap / car can be dropped where it is needed. Accept: its strength can be changed. Do not accept 'it is stronger' on its own.
Notes / working: An electromagnet is only magnetic when a current flows. The crane operator can **switch it off** to drop the car in the right place. A permanent magnet could not let go.

(e) What is the independent variable in Sami's investigation? **[1]**

Answer: current or the current or electric current or size of the current or the size of the current or current in A or the current in the wire
Notes / working: The independent variable is the one Sami changes on purpose: the **current**. The number of nails lifted is the dependent variable.

(f) One result is anomalous. At what current was this result? **[1]**

Answer: 0.8 A
Notes / working: The number of nails goes up by 3 each time the current goes up by 0.2 A (3, 6, 9, …, 15), so at 0.8 A it should be about 12. The result of 5 at **0.8 A** does not fit the pattern: it is anomalous.

(g) Predict how many nails the electromagnet would lift with a current of 1.2 A. **[1]**

Answer: 18 nails
Notes / working: The pattern is +3 nails for each extra 0.2 A, so 15 + 3 = **18** nails (accept 17 to 19).

(h) Describe the pattern in Sami's results. Ignore the anomalous result. **[1]**

Answer: 1 mark: as the current increases, the number of nails lifted increases. Accept: the number of nails goes up by 3 for every 0.2 A / is proportional to the current / doubles when the current doubles. Do not accept a list of the numbers with no pattern.
Notes / working: The **bigger the current, the more nails** the electromagnet lifts: about 3 more nails for each extra 0.2 A, so doubling the current doubles the number of nails.

(i) Give **one** control variable Sami must keep the same. **[1]**

Answer: 1 mark for any one: number of turns (of wire) in the coil; the same iron nail / core; the same size / type / mass of steel nails; the same type / thickness of wire. Do not accept 'the current' (this is the independent variable) or 'the same equipment' on its own.
Notes / working: For example, the **number of turns** of wire (20) or the same iron core. Each of these also changes the strength of an electromagnet, so it must stay the same for a fair test.

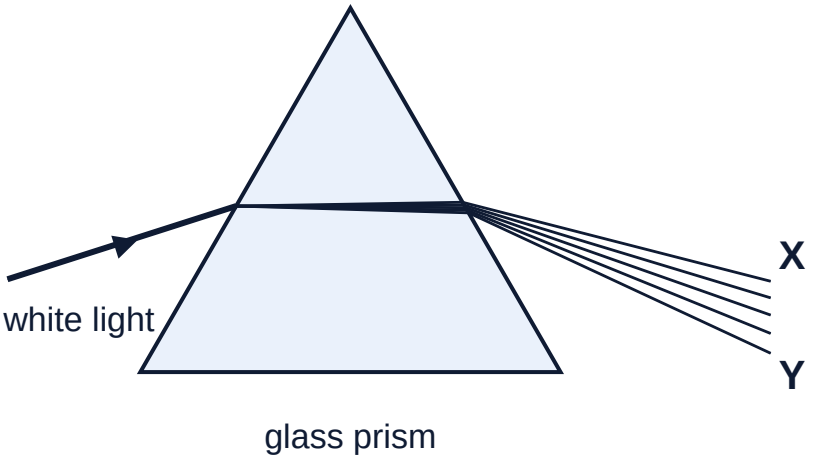
(j) Sami took each reading only once. Suggest how he could make his results more reliable, and explain why this helps. **[1]**

Answer: 1 mark: repeat each reading (at least 2 more times) and calculate a mean, so anomalous results can be spotted / ignored. Both the action and a reason are needed for the mark.
Notes / working: Repeat each reading three times and work out the **mean**. An anomalous result like the one at 0.8 A would then be easy to spot and leave out, so the results are more reliable.

(k) The Earth also has a magnetic field. Complete the sentence with one word. **[1]**
The Earth has a magnetic field because its _____ acts like a magnet.

Answer: core or iron core or outer core or inner core or metal core
Notes / working: The Earth's **core** is made mostly of iron and nickel, and it acts like a huge magnet. This is why a compass points north.

5 This question is about light and colour: a rainbow and the coloured lights on a school stage. **[1]**



(a) A rainbow forms when sunlight passes through raindrops. In the laboratory, Hana shines a ray of white light into a glass prism (diagram not drawn to scale). The white light splits into colours. What is the name for the splitting of white light into colours?

Answer: dispersion or dispersal
Notes / working: White light is a mixture of many colours. The prism slows and bends each colour by a slightly different amount, so the colours spread out: this is **dispersion**.

(b) Look at the diagram of Hana's prism. Which ray, X or Y, is red light? **[1]**

Answer: X
Notes / working: Red light is refracted (bent) the **least** and violet light the most. X has changed direction least, so **X is red** and Y is violet.

(c) Explain why the light changes direction when it goes from the air into the glass. **[1]**

Answer: 1 mark: light slows down / changes speed when it enters the glass (glass is denser than air). Do not accept 'the glass reflects it'.
Notes / working: Light travels more **slowly** in glass than in air. When a ray enters the glass at an angle, the change of speed makes it change direction: this is **refraction**.

(d) At a school concert, the stage is lit with coloured spotlights. A red spotlight and a green spotlight shine onto the same part of a white screen. What colour is seen where they overlap? **[1]**

Answer: yellow
Notes / working: Adding coloured lights: red + green = **yellow**. (Red + green + blue = white.)

(e) On the stage, a singer wears a white shirt and blue trousers. Only a red spotlight shines on him. Describe and explain what colour his shirt and his trousers look. **[2]**

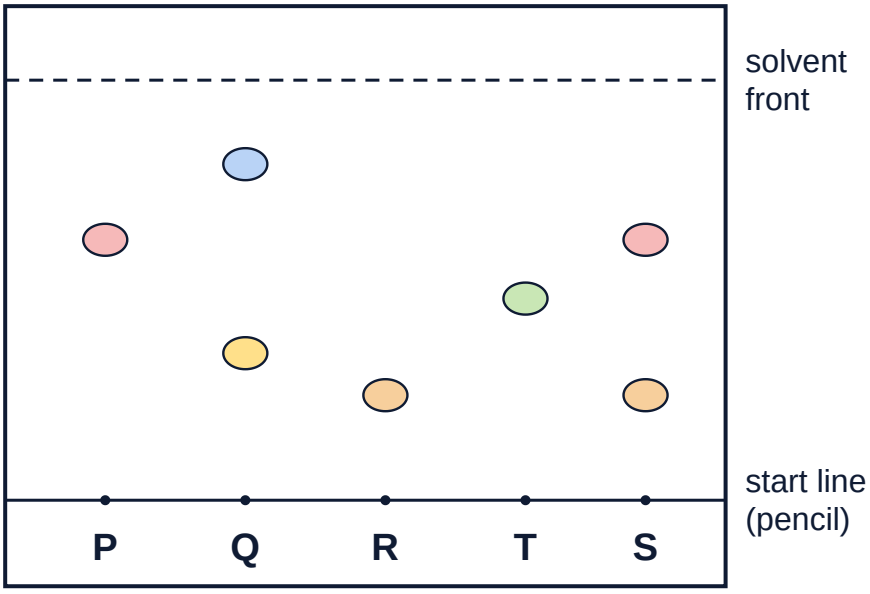
> **Answer:** 1 mark: the shirt looks red because white reflects all colours, so it reflects the red light. 1 mark: the trousers look black because blue (objects) absorb red light / reflect only blue light, and there is no blue light to reflect. Colour without a reason gets no mark.
> **Notes / working:** A white object reflects every colour that falls on it, so the shirt reflects the red light and **looks red**. Blue trousers reflect only blue light and absorb the other colours. There is no blue in the red spotlight, so the red is absorbed, no light is reflected and the trousers **look black**.

(f) A spotlight beam hits a flat mirror on the stage. The angle of incidence is 25°. What is the angle between the incident ray and the reflected ray? **[1]**

> **Answer:** 50 °
> **Notes / working:** Law of reflection: angle of reflection = angle of incidence = 25°. Both angles are measured from the normal, on opposite sides, so the angle between the rays is 25° + 25° = **50°**.

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6 In the chemistry laboratory, Laila tests the colouring in an orange sports drink, S. She uses paper chromatography to compare it with four food dyes, P, Q, R and T. **[1]**



(a) Which **two** food dyes are in the sports drink S? Write two letters.

> **Answer:** P and R or PR or RP or R and P or P,R or R,P or P R or R P or P & R or R & P
> **Notes / working:** S has two spots. One is at the same height as the spot of P and the other at the same height as the spot of R, so S contains **P and R**.

(b) Which food dye, P, Q, R or T, is a mixture of more than one substance? **[1]**

> **Answer:** Q
> **Notes / working: Q** separates into two spots, so it contains two different substances. P, R and T each give one spot.

(c) Laila also makes some copper sulfate crystals. Her 20 g sample contains 18.4 g of copper sulfate. The rest is water and unreacted copper oxide. Calculate the percentage purity of Laila's copper sulfate. Show your working. **[2]**

> **Answer:** 92 %
> **Notes / working:** percentage purity = mass of the pure substance ÷ total mass × 100 = 18.4 ÷ 20 × 100 = **92%**. (M1 for 18.4 ÷ 20 (× 100); A1 for 92.)

(d) Laila's copper sulfate is not pure. Suggest **one** reason why the product of a chemical reaction is often not pure. **[1]**

> **Answer:** 1 mark for any one: some reactant is left over / unreacted (e.g. copper oxide); more than one product is made / other (side) reactions happen; the reactants were not pure; the product is mixed with the solvent (water) / dirt from the apparatus.
> **Notes / working:** For example, some of a reactant does not react and stays mixed with the product (like the unreacted copper oxide), or the reaction makes more than one product, so the product is an **impure mixture**.

(e) Laila makes two sugary sports drinks. **[1]**

Drink	Mass of sugar in g	Volume of drink in cm^3
A	30	500
B	20	250

Which drink, A or B, is more concentrated?

> **Answer:** B
> **Notes / working:** Compare the mass in the same volume. A: 30 g in 500 cm^3 = 6 g in 100 cm^3. B: 20 g in 250 cm^3 = 8 g in 100 cm^3. **B** has more sugar particles in each cm^3, so it is more concentrated.

(f) The school nurse keeps instant cold packs for sports injuries. When a cold pack is squeezed, a solid dissolves in the water inside. The temperature falls from 21 °C to 4 °C. Is dissolving this solid exothermic or endothermic? **[1]**

> **Answer:** endothermic or endothermic process or endothermic reaction or it is endothermic
> **Notes / working:** The temperature **falls** (by 17 °C), so energy is taken in from the surroundings: the process is **endothermic**.

(g) The table shows the solubility of potassium nitrate at different temperatures. **[2]**

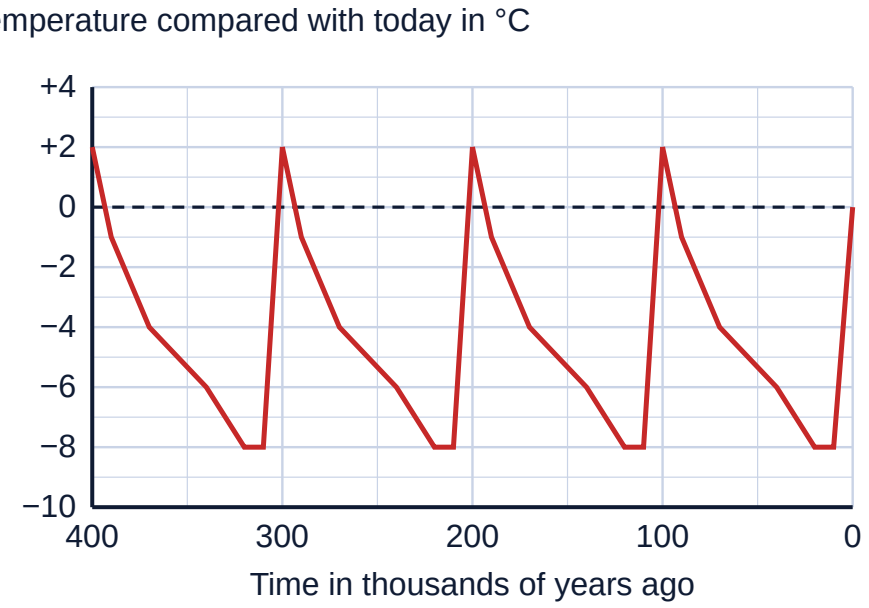
Temperature in °C	0	20	40	60	80
Solubility in g per 100 g of water	13	32	64	110	169

Laila makes a saturated solution of potassium nitrate in 100 g of water at 60 °C. She cools it to 20 °C. What mass of crystals forms? Show your working.

Answer: 78 g
Notes / working: At 60 °C, 110 g dissolves in 100 g of water. At 20 °C only 32 g can stay dissolved. The rest comes out as crystals: 110 − 32 = **78 g**. (M1 for 110 and 32 read from the table or 110 − 32; A1 for 78.)

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7 The graph shows how the average temperature of the Earth changed over the last 400 000 years, compared with the temperature today. The pattern is based on evidence from ice cores. **[1]**



(a) What is the lowest temperature shown on the graph, compared with today?

Answer: -8 °C
Notes / working: The lowest points of the line are at **−8 °C**, which means 8 °C colder than today. These cold times are ice ages.

(b) The climate goes through a cycle of ice ages and warm periods. How many thousand years are there between one warm period and the next? **[1]**

Answer: 100 thousand years
Notes / working: Warm peaks are at 400, 300, 200 and 100 thousand years ago (and today), so the cycle repeats every **100 thousand years** (100 000 years). Climate cycles take place over very long time periods.